

How a magnetic flux tube changes across neighbouring sections

Three tensor parts. Each column isolates one positive coefficient.

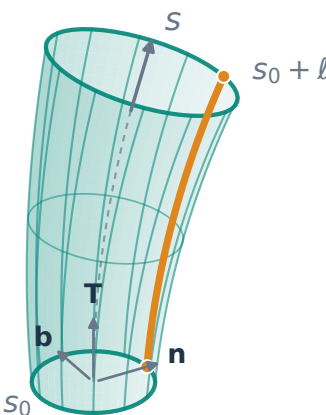
$$\mathbf{T} = \mathbf{B}/B, \quad \mathbf{b} = \mathbf{T} \times \mathbf{n}$$

$$M = \begin{bmatrix} a & q \\ p & d \end{bmatrix} = \frac{\theta}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} \delta_g & \beta_g \\ \beta_g & -\delta_g \end{bmatrix} + \frac{\alpha}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

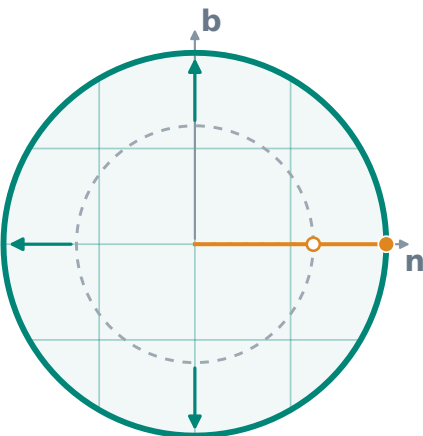
ISOTROPIC DILATION

Uniform expansion

$$\theta > 0$$



Normal-plane view



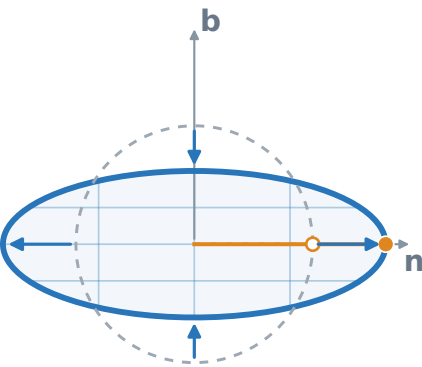
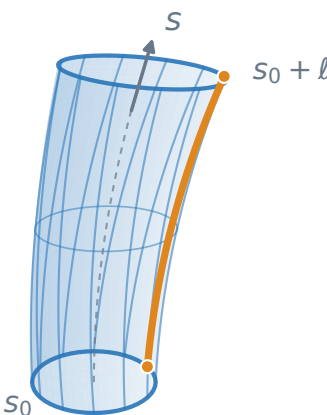
$$\theta = a + d$$

$$A_1/A_0 = e^{\theta l}$$

SYMMETRIC STRAIN / BIAxIAL SPLAY

Stretch along n; squeeze along b

$$\delta_g > 0$$

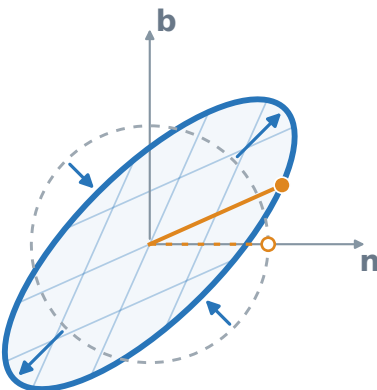
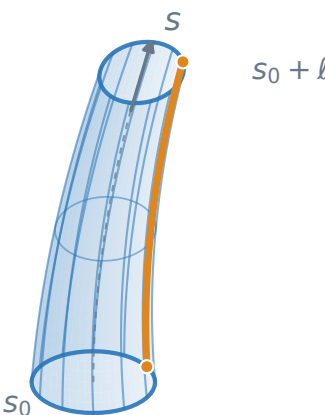


$$\delta_g = a - d$$

$$A_1 = A_0$$

Stretch and squeeze at 45 degrees

$$\beta_g > 0$$



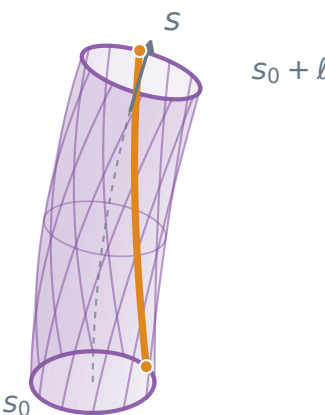
$$\beta_g = p + q = D/B$$

$$A_1 = A_0$$

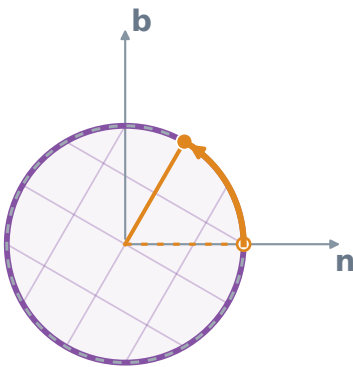
LOCAL ROTATION

Rotate marked field lines

$$\alpha > 0$$



n right, b up; T points towards the viewer



$$\alpha = p - q = \mu_0 j_{\parallel} / B$$

$$\Delta\phi = \alpha l / 2, \quad A_1 = A_0$$

A_0, A_1 : areas normal to $\mathbf{T}$ at s_0 and $s_0 + l$; l : arc length between sections along the reference line.

Dashed grey: initial section Colour: mapped section Orange: the same labelled field line

Negative signs reverse the illustrated map.

$$\xi(s_0 + l) = \exp(lM) \xi(s_0)$$

Constant local coefficients; transported frame (torsion = 0 here). s is distance, not time.